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in

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Submitted by

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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

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This is to certify that the work entitled “**NATIONAL SERVICE SCHEME (BNSK359)**” is a bonafide work carried out by **CHANJALA K P (1VI24CS018)**, for the fulfillment of the subject **NSS(BNSK359)** in 3rd Semester of **Visvesvaraya Technological University, Belagavi** during the academic year 2025-2026. It is certified that all corrections/suggestions indicated for internal assessment have been incorporated in the report. The report has been approved as it satisfies the academic requirements in respect of the work prescribed for the said degree.

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DECLARATION

I, Chanjala K P bearing the USN: **1VI24CS018** student of Bachelor of Engineering in Dept. of COMPUTER SCIENCE AND ENGINEERING at Vemana institute of Technology, Bengaluru, hereby declare that the **NON CREDIT MANDATORY COURSE- “NATIONAL SERVICE SCHEME (BNSK359)”** work has been carried out by me under the supervision and guidance of Mrs. R Shwetha **Assistant Professor**, Dept. of CSE, submitted by me as a partial fulfillment for the award of Bachelor of Engineering degree in Computer Science and Engineering from Visvesvaraya Technological University, Belagavi during 2025-26. I also declare that, this work has not been submitted previously for the award of any degree or diploma, by me, to any institution.

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ABSTRACT

Indian agriculture plays a vital role in the economic and social development of the country. With growing concerns about environmental sustainability, soil health, and farmer welfare, organic farming has gained importance as an eco-friendly and sustainable agricultural practice. This NSS activity was organized to provide students with practical exposure to organic farming, understand the evolution of Indian agriculture from traditional to modern practices, and explore the importance of marketing connectivity for farmers. As part of the activity, I visited a tissue culture plantation and the Krishi Mela at GKVK College. The visit helped me to gain firsthand knowledge about tissue culture techniques, organic farming methods, modern agricultural technologies, and various government initiatives supporting farmers. The activity enhanced awareness about sustainable agriculture, improved understanding of agricultural marketing systems, and encouraged students to appreciate the role of innovation and technology in shaping the future of Indian agriculture.

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CHAPTER 1

INTRODUCTION TO ORGANIC FARMING

Organic farming is an environmentally friendly method of agriculture that focuses on growing crops using natural inputs and sustainable practices. It avoids the use of synthetic fertilizers, chemical pesticides, and genetically modified organisms, thereby protecting soil health, water resources, and biodiversity. Organic farming promotes the use of organic manures such as compost, green manure, and bio-fertilizers, which help maintain soil fertility and improve crop productivity in a natural way.

In traditional Indian agriculture, organic farming practices were widely followed before the introduction of chemical-based farming. Over time, the excessive use of chemicals led to soil degradation, water pollution, and health-related concerns. Organic farming has re-emerged as a solution to these problems, ensuring safe food production while preserving the ecological balance.

Organic farming not only benefits the environment but also supports the well-being of farmers and consumers. It reduces input costs, improves soil quality, and produces nutritious, chemical-free food. With increasing awareness, government support, and market demand for organic products, organic farming is playing an important role in achieving sustainable agricultural development in India.

The growing imbalance between freshwater availability and demand results in water scarcity, a critical global challenge. This scarcity not only threatens ecosystems but also impacts human health, food security, and economic stability. Therefore, effective water conservation—the sustainable management and protection of water resources to meet current and future needs—is no longer an option but an urgent necessity.

Conservation involves minimizing wastage, improving efficiency of use, protecting water quality, and ensuring equitable access for all.

CHAPTER 2

IMPORTANCE OF NSS ACTIVITY

Introduction to NSS

The National Service Scheme (NSS) is a youth-centric program launched by the Government of India with the motto “*Not Me, But You.*” It aims to develop the personality and character of students through voluntary community service. NSS provides students with opportunities to work for the welfare of society while gaining valuable life experiences. Through NSS activities, students are encouraged to understand social issues and actively contribute to nation-building.

Social Awareness and Community Development

One of the major importance of NSS activities is creating social awareness among students. NSS exposes students to real-life social problems such as poverty, illiteracy, unemployment, environmental degradation, and challenges faced by rural communities. Through community outreach programs, village visits, and awareness campaigns, students gain firsthand experience of societal conditions. This helps them develop empathy, compassion, and a sense of responsibility toward society.

Holistic Personality Development

NSS plays a significant role in the holistic development of students. Participation in NSS activities helps students develop leadership qualities, teamwork, discipline, and effective communication skills. Volunteers learn to work collectively, respect diverse opinions, and manage responsibilities. These skills help in shaping students into confident individuals who are capable of handling real-world challenges beyond academics.

Experiential and Practical Learning

NSS promotes learning through experience rather than theoretical knowledge alone. Activities such as field visits, study tours, and interaction with experts provide practical exposure to students. In agriculture-related NSS activities, students learn about farming practices, sustainable development, and modern agricultural technologies. This experiential learning helps students understand concepts more effectively and relate academic knowledge to practical applications.

Environmental Awareness and Sustainability

Environmental conservation is an important focus area of NSS activities. Programs related to tree plantation, organic farming, cleanliness drives, water conservation, and waste management create awareness about environmental protection. NSS encourages students to adopt eco-friendly practices and spread awareness among the public. Such activities play a crucial role in promoting sustainable development and protecting natural resources for future generations.

Contribution to Rural and Agricultural Development

NSS activities contribute significantly to rural development. Through village adoption programs, agricultural visits, and Krishi Melas, students gain exposure to rural life and agricultural practices. Activities related to organic farming, tissue culture plantations, and farmer interaction help students understand the importance of sustainable agriculture and marketing connectivity. These initiatives create awareness about improving farmers' income and strengthening the agricultural sector.

National Integration and Social Responsibility

NSS promotes national integration by bringing students from different backgrounds together for a common cause. Participation in NSS activities helps students understand the importance of unity, cooperation, and collective effort in nation-building. NSS instills values such as patriotism, selflessness, and social responsibility, encouraging students to contribute positively to the development of the nation.

Strengthening Institution–Community Relationship

NSS acts as a bridge between educational institutions and society. Through outreach programs, awareness campaigns, and study visits, NSS ensures that institutions actively participate in community development. Interaction with farmers, scientists, and experts during NSS activities enhances students' knowledge and respect for grassroots-level work. This strengthens the bond between students and society.

PROBLEM STATEMENT

Background of the Problem

Indian agriculture has undergone significant transformation over the years, shifting from traditional and nature-based farming practices to modern, technology-driven methods. While these advancements have increased productivity, they have also led to several challenges such as excessive dependence on chemical fertilizers and pesticides, degradation of soil fertility, environmental pollution, and health-related concerns. These issues have raised serious questions about the sustainability of present-day agricultural practices.

Challenges in Organic Farming Adoption

Although organic farming offers a sustainable and eco-friendly alternative, its adoption among farmers remains limited. Many farmers lack proper awareness, technical knowledge, and training related to organic farming practices. The initial transition period from conventional farming to organic farming is challenging due to lower yields and certification requirements. Limited access to organic inputs and insufficient institutional support further discourage farmers from adopting organic methods.

Issues in Agricultural Awareness and Education

A major problem in Indian agriculture is the lack of awareness among the younger generation and students regarding agricultural practices and rural challenges. Urbanization and modernization have reduced direct interaction between students and farming communities. As a result, there is limited understanding of traditional agricultural knowledge, sustainable farming methods, and the importance of agriculture in national development.

Marketing Connectivity Problems for Farmers

One of the most critical issues faced by farmers is poor marketing connectivity. Farmers often depend on middlemen to sell their produce, which results in low profit margins and exploitation. Lack of access to organized markets, price information, storage facilities, and digital platforms restricts farmers from reaching consumers directly. Even organic farmers face difficulties in finding reliable markets and fair prices for their produce.

Technological and Resource Constraints

Although modern agricultural technologies such as tissue culture, precision farming, and digital platforms exist, many farmers are unable to access or implement them effectively. High costs, lack of technical support, and limited infrastructure in rural areas act as major barriers. Small and marginal farmers especially struggle to adopt advanced technologies due to financial and resource constraints.

Need for Student and Institutional Involvement

Educational institutions and students have a significant role to play in addressing agricultural challenges. However, there is a gap between academic learning and real-world agricultural issues. Limited exposure to practical farming environments prevents students from understanding ground-level problems faced by farmers. NSS activities can serve as an effective medium to bridge this gap by promoting awareness, education, and community engagement.

Relevance of the Study

In view of the above challenges, there is a strong need to create awareness about sustainable agriculture, organic farming practices, and improved marketing connectivity. The NSS study visit to a tissue culture plantation and Krishi Mela at GKVK College was undertaken to address these issues by providing practical exposure to students. The activity aimed to enhance understanding of agricultural innovations, promote sustainable practices, and highlight the importance of connecting farmers to market.

CHAPTER 4

OBJECTIVE OF THE STUDY / ACTIVITY

Primary Objective

The primary objective of this NSS activity was to provide students with practical exposure to organic farming practices and to understand the evolution of Indian agriculture from traditional methods to modern, technology-based approaches. The activity aimed to bridge the gap between theoretical knowledge and real-world agricultural practices through field visits and direct interaction with experts and farmers.

Understanding Organic Farming Practices

One of the key objectives of the activity was to create awareness about organic farming and its importance in sustainable agricultural development. The visit aimed to help students understand the principles of organic farming, including the use of natural inputs, soil health management, and environmentally friendly farming techniques. This objective also focused on highlighting the benefits of organic farming for farmers, consumers, and the environment.

Study of Indian Agriculture: Past, Present, and Future

Another important objective was to study the transformation of Indian agriculture over different periods. The activity aimed to help students understand traditional agricultural practices followed in the past, the present-day use of modern technology and machinery, and future trends such as tissue culture, precision farming, and smart agriculture. This understanding was essential to appreciate the continuous development of the agricultural sector.

Exposure to Modern Agricultural Technologies

The visit to the tissue culture plantation was organized with the objective of familiarizing students with advanced agricultural technologies. Through this exposure, students were expected to learn about tissue culture techniques, their role in producing disease-free and high-quality plants, and their contribution to improving crop productivity and food security.

Awareness on Marketing Connectivity for Farmers

A major objective of the NSS activity was to understand the importance of marketing connectivity in agriculture. The visit to the Krishi Mela aimed to educate students about various marketing channels, government initiatives, digital platforms, and farmer-support systems that help farmers sell their produce at fair prices. This objective focused on understanding how improved market access can enhance farmers' income and reduce dependency on middlemen.

Interaction with Farmers and Agricultural Experts

The activity aimed to encourage interaction between students, farmers, and agricultural scientists. Such interactions were intended to provide insights into real-life challenges faced by farmers, innovative solutions adopted in agriculture, and the role of research institutions in supporting the farming community. This objective helped students gain practical knowledge beyond classroom learning.

Promoting Social Responsibility and NSS Values

Another objective of the activity was to instill the values of social responsibility, service, and community engagement among NSS volunteers. By participating in agriculture-related activities, students were encouraged to develop empathy towards farmers and rural communities and understand their role in national development.

Encouraging Youth Participation in Sustainable Development

The NSS activity also aimed to motivate students to actively participate in promoting sustainable agricultural practices. By creating awareness about organic farming and environmental conservation, the activity sought to inspire students to contribute towards sustainable development and support eco-friendly initiatives in the future.

METHODOLOGY / DESCRIPTION OF THE ACTIVITY

Planning and Preparation

The NSS activity was carefully planned in advance with the objective of providing practical exposure to organic farming, modern agricultural technologies, and marketing connectivity for farmers. As part of this initiative, the tissue culture plantation visit was planned and organized by the students to **V2 Plants Inc** and a visit to **Krishi Mela at GKVK** was conducted. Necessary permissions were obtained, and the activity schedule was prepared. NSS volunteers were briefed about the purpose of the visit, objectives of the study, and the discipline to be maintained during the activity.



Selection of Study Locations

Two important study locations were selected to achieve the objectives of the activity. **V2 Plants Inc.** was chosen to understand tissue culture techniques and modern plant propagation methods, while **GKVK College** was selected for visiting the Krishi Mela to gain exposure to organic farming practices, agricultural innovations, and farmer marketing initiatives. These locations provided a balanced understanding of both technological and practical aspects of Indian agriculture.

Visit to Tissue Culture Plantation at V2 Plants Inc.

As part of the NSS activity, the I visited the tissue culture plantation at **V2 Plants Inc.**, which was planned by NSS team members. During the visit, experts explained the complete process of tissue culture, including selection of plant material, sterilization, multiplication, rooting, and hardening of plants. I observed the laboratory setup and learned how disease-free and high-quality planting materials are produced in a controlled environment. The visit helped me to understand the importance of tissue culture in improving crop productivity, ensuring uniform quality, and supporting sustainable and organic farming practices. It also highlighted the role of biotechnology in the future development of Indian agriculture. During the visit to V2 Plants Inc., I interacted with technical staff and experts who shared their knowledge on modern agricultural technologies, challenges in plant production, and the advantages of tissue culture methods. These interactions provided valuable insights into the practical application of scientific research in agriculture and helped me relate theoretical concepts to real-world practices.



Visit to Krishi Mela at GKVK College

The NSS volunteers also visited the **Krishi Mela organized at GKVK College**. The Krishi Mela featured various stalls displaying organic products, traditional seeds, bio-fertilizers, bio-pesticides, and modern agricultural equipment. Demonstrations and exhibitions showcased innovative and eco-friendly farming techniques. The visit helped students understand how traditional knowledge and modern technology can work together in agriculture.



Study of Marketing Connectivity for Farmers

Special emphasis was given to understanding marketing connectivity during the Krishi Mela. Volunteers learned about Farmer Producer Organizations (FPOs), government schemes, e-NAM platforms, and direct farmer-to-consumer marketing models. These initiatives highlighted ways to improve farmers' income, reduce dependence on middlemen, and ensure fair pricing of agricultural produce.

Data Collection and Observation

Information for the study was collected through direct observation, interaction with experts and farmers, discussions at stalls, and review of informational materials displayed during the visits. NSS volunteers noted key points related to organic farming practices, tissue culture techniques, and agricultural marketing systems. These observations helped in understanding the present agricultural scenario and future opportunities.

NSS CHAPTER 6 STAGES IN TISSUE CULTURE OBSERVED AT V2 PLANTS INC.

Tissue culture is a modern plant propagation technique used for the large-scale production of healthy and disease-free plants under controlled laboratory conditions. During the NSS study visit to **V2 Plants Inc., Anekal**, the volunteers observed the complete tissue culture process, starting from laboratory preparation to nursery-level hardening. The plantation mainly focuses on the large-scale production of **banana plants and ornamental plants**, which are in high demand for commercial cultivation and landscaping purposes. The visit provided practical exposure to advanced agricultural biotechnology techniques.



Media Preparation

The first stage observed was media preparation. In this stage, a nutrient medium is prepared by mixing water, sugars, minerals, vitamins, and plant growth regulators. This medium provides essential nutrients required for plant growth. At V2 Plants Inc., the prepared medium was poured into culture bottles and sealed properly before sterilization. Different media formulations are used depending on whether the plants are **banana varieties or ornamental plants**. After pouring the prepared culture medium into the glassware, sterilization was carried out by subjecting it to a **specific temperature and pressure** to eliminate microorganisms and prevent contamination. The prepared culture medium was then transferred to the next section through a **pass box** to maintain sterile conditions.



Inoculation

After sterilization, the explant is transferred to the prepared culture medium. This process is called inoculation and is carried out inside a laminar airflow chamber to maintain aseptic conditions. At V2 Plants Inc., trained technicians performed inoculation carefully to avoid contamination and ensure proper placement of the explant in the medium.



Multiplication Stage

In this stage, the explant starts producing multiple shoots. Plant growth regulators in the culture medium help in rapid multiplication. It was observed that this stage allows large numbers of plants to be produced from a single explant within a short period. This makes tissue culture an efficient method for mass propagation.



Rooting Stage

Once sufficient shoots are formed, they are transferred to another medium that promotes root development. This stage is known as the **rooting stage**. At V2 Plants Inc., volunteers observed how roots were induced to develop, preparing the plantlets for further growth. Proper rooting is essential for the survival of plants when transferred to natural conditions.



Cutting and Separation of Plantlets

After rooting, the plantlets are carefully separated or cut into individual units. This stage requires precision to avoid damage to the roots and shoots. The separated plantlets are then prepared for the next stage of growth. This step increases the number of plants ready for hardening.

Storage and Growth Room Observation

The volunteers also visited the **storage and growth rooms**, where controlled temperature, humidity, and light conditions are maintained. These rooms provide an ideal environment for the growth and development of tissue-cultured plants. Shelves with culture bottles were observed, indicating systematic organization and monitoring.



Hardening Process

Hardening is the process of gradually acclimatizing tissue-cultured plants to external environmental conditions. The plants are transferred from the laboratory to greenhouses or shade nets. This step strengthens the plantlets and improves their survival rate when planted in soil.



Final Transfer to Field or Nursery

After successful hardening, the plants are ready to be transferred to nurseries or agricultural fields. These plants are healthy, uniform, and disease-free, making them suitable for commercial cultivation and organic farming practices.



OBSERVATIONS AND FINDINGS

Observations from the Tissue Culture Plantation at V2 Plants Inc.

During the visit to the tissue culture plantation at **V2 Plants Inc.**, it was observed that tissue culture is an efficient and scientific method of plant propagation. The process is carried out in a controlled and sterile environment, which helps in producing disease-free and genetically uniform plants. The use of advanced laboratory facilities and skilled technical staff ensures high survival rates and consistent plant quality. It was also observed that tissue culture plants require less time for growth compared to conventional propagation methods. This technique supports large-scale production of planting material and plays an important role in meeting agricultural demands. The visit highlighted the significance of biotechnology in modern agriculture and its potential to improve productivity and sustainability.



Role of Tissue Culture in Sustainable Agriculture

The findings revealed that tissue culture technology contributes significantly to sustainable agriculture. By producing healthy and high-quality planting materials, tissue culture reduces crop loss and increases yield. It also supports organic farming by minimizing the need for chemical treatments to control diseases. This technique is especially useful for horticultural crops, medicinal plants, and commercial plantations.



Key Findings of the Study

- Tissue culture technology enables rapid and disease-free plant production.
- Organic farming promotes soil health and environmental sustainability.
- Krishi Melas are effective platforms for farmer education and market awareness.
- Marketing connectivity is essential for improving farmers' income.
- NSS activities enhance practical learning and social responsibility among students.



CHAPTER 8

CONCLUSION

The NSS study visit to the tissue culture plantation at **V2 Plants Inc.** and the **Krishi Mela** at **GKVK College** proved to be a valuable and enriching learning experience for the volunteers. The activity provided practical exposure to organic farming practices, modern agricultural technologies, and the changing trends in Indian agriculture. Through the visit to the tissue culture plantation, students gained insight into advanced plant propagation techniques and their role in producing healthy, disease-free planting material. The Krishi Mela helped volunteers understand organic farming methods, sustainable agricultural practices, and various government initiatives aimed at supporting farmers. The activity also highlighted the importance of marketing connectivity, digital platforms, and direct market access in improving farmers' income and reducing dependence on intermediaries. Interaction with experts, scientists, and farmers enhanced students' understanding of real-life agricultural challenges and solutions. Overall, the NSS activity successfully fulfilled its objectives by promoting experiential learning, environmental awareness, social responsibility, and appreciation for sustainable agricultural development, thereby encouraging students to contribute meaningfully to rural and national progress.

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